Abu Jafar Saifullah

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Experience

Optimizely, Content Marketing Platform — Dhaka, Bangladesh
Software Engineer II (promoted from SE I, Apr 2026) — *Apr 2024 – Present, Full-time*

- **Design System Migration:** Contributed to the platform's multi-year migration to **Axiom** (`@optiaxiom/react`, public design system). Landed **26 merged PRs** in the open-source repo — V1→V2 component conversions plus net-new components (Aside, SubNav, Range) — and **10+ adapter components** in the product. Wrote the Migration Guide and Icon Contribution Guide used by other product teams.
- **Axiom V3 Brand Refresh:** Worked on the V3 token/color rollout in the product, including a **module-federation singleton fix** that eliminated duplicate React-context errors across micro-frontends.
- **AI SDK Integration:** Worked on end-to-end integration of an internal AI SDK. Migrated the backing service from **Express to Hono**, set up the **AWS CodeArtifact** publishing pipeline, built the permissions layer and test suite, and wrote the rollout documentation.
- **AI Event Instrumentation:** Built a reusable event-tracking React hook and instrumented structured AI-context events across **10+ product surfaces**, enabling downstream AI agents to consume product context.
- **Block-based AI Document UI:** Implemented backend block-context extraction for LLM prompts, frontend block-form interaction, and AI resource rendering for the platform's AI document interface (Adaptive-Cards-style pattern).
- **Spotlight Global Search:** Contributed to the platform's **Spotlight** global search with status pills, keyboard activation (`Cmd+K`), search-result separation, and segment-event telemetry. Built during, and won, the **Optimizely Global Hackathon 2025** as part of a team.
- **Dark Mode:** Implemented **Dark Mode** across **15+ surfaces** (modals, dropdowns, tables, library, notifications, plan share, asset upload); refactored to **CSS variables** and prevented bleed-through into legacy UI.
- **Platform Quality:** Resolved duplicate comment/task creation on rapid clicks (server + client), fixed duplicate in-app notifications, restored thread-reply notifications to original commenters, and added **8 missing i18n locale variants**.
- **Code Review:** Reviewed **300+ pull requests** across product client/server and AI services; participated in sprint planning and technical design discussions.

Orion Informatics Ltd. — Dhaka, Bangladesh
Software Engineer — *Jan 2023 – Jan 2024, Full-time*

- Built custom graphs and advanced data filters for an Attendance Management System to enhance reporting capabilities.
- Built a two-service **ASP.NET Core** webhook relay that ingests TradeStation alerts, validates request integrity, and routes by trade type to a file-based queue consumed by an automated trading bot; deployed on **Azure App Service**.

SEMERU Lab, William & Mary — Remote
Research Intern, supervised by Dr. Denys Poshyvanyk — *Aug 2023 – Dec 2023*

- Built a programming language-specific tokenizer to improve NLP model predictions for software engineering tasks.

Education

University of Dhaka — Dhaka, Bangladesh
M.Sc. in Software Engineering (completed; final result pending); ***GPA: 3.92/4.00*** — *Apr 2025 – Sep 2026*
B.Sc. in Software Engineering; ***GPA: 3.89/4.00*** — *Jan 2019 – Dec 2023*

Notre Dame College — Dhaka, Bangladesh
Higher Secondary Certificate; ***GPA: 5.00/5.00*** — *Jun 2016 – Jun 2018*

Publications

Abu Jafar Saifullah, Tasmia Zerin, Zerina Begum, Kazi Sakib. *"NavA11y: A Dynamic Analysis Approach for WCAG 2.4 Focus-Behavior Evaluation."* In *Proceedings of the 21st International Conference on Evaluation of Novel Approaches to Software Engineering* **(ENASE 2026)**, vol. 2, pp. 1019–1030. SCITEPRESS. Benidorm, Spain. *Full paper; presented in person, May 2026.* DOI: 10.5220/0015045700004015 [Paper] [Code]

Projects

Blueberry Auto | *React 18, TypeScript, Vite, Hono, Supabase, Tailwind, shadcn/ui, pnpm workspaces, Vercel* | Live | *Lead Dev*

- Vehicle auction and marketplace platform; pnpm monorepo with a Vite/React frontend, Hono API, Supabase backend, and a separate crawler app. Leading end-to-end build (architecture, OAuth, Supabase migrations, edge functions, Vercel deploy) as the contracted developer; **150+ merged PRs**.

Opti Retreat Voting System | *React, TypeScript* | GitHub | Live | *Optimizely Retreat 2024*

- Internal voting application built for Optimizely's annual company retreat; React + TypeScript SPA deployed on Netlify, used by the Bangladesh office.

NavA11y | *Node.js, Playwright, Chromium* | Code | *ENASE 2026*

- Dynamic WCAG 2.4 accessibility testing tool, the first to cover all five focus-behavior success criteria. Drives a real browser to simulate keyboard navigation and inspect rendered page state at each focus event. Achieved 100% precision and recall on a 22-page labeled dataset; detected 2,947 violations across 26 production websites with 90% confirmed true-positive rate.

RepairA11y | *Node.js, Playwright, Vitest* | *M.Sc. Thesis, 2026*

- LLM-driven automated repair pipeline for WCAG 2.4 accessibility violations; evaluates whether runtime-evidence prompting outperforms static-evidence (HTML/screenshot) prompting for accessibility patch generation. Uses NavA11y as validation oracle.

Clippy | *Angular, Node.js, AWS S3* | GitHub | *ICSE SCORE 2023, Top 6*

- Smart PDF reader with knowledge graph extraction, citation network visualization, and paper recommendation. OTP-based auth, Azure CI/CD; Top 6 finalist in the ICSE 2023 Student Contest on Software Engineering (SCORE).

Mr. Bin | *MERN, Spring Boot 3, TensorFlow, Maps API* | *JavaFest 2023 (Therap)*

- Full-stack recycling and exchange marketplace with ML-based object detection for item classification; MERN stack with a Spring Boot 3 / Java 17 backend rewrite. Entered in JavaFest 2023 (Therap competition).

API Miners GPT | *ASP.NET Core, React, OpenAI API, EF Core* | *BUET Hackathon, Best UI/UX*

- GPT-powered PDF chatbot with voice input, OCR, and Stable Diffusion image generation; .NET backend with SQL Server, React + Tailwind CSS frontend. Won Best UI/UX at BUET Hackathon 2023.

MiniFacebook | *React, Node.js, Docker, NGINX* | GitHub

- Demonstrated monolith-to-microservices decomposition by splitting a Node.js app into User, Story, and Status services behind an NGINX reverse proxy / load balancer, orchestrated with Docker Compose.

BPE Tokenizer | *Python, Flask, HuggingFace* | PyPI | *Published Package*

- Byte Pair Encoding tokenizer implemented from scratch and published as `python-bpe-tokenizer` on PyPI; includes a Flask web interface for training on custom corpora and tokenize/detokenize workflows.

EmitXplorer | *React, ASP.NET Core, Python/ML, TensorFlow* | *NASA Space Apps 2023*

- Gamified greenhouse gas visualization platform using SCISAT satellite data and ML-based CO_2 concentration prediction; built for NASA Space Apps Challenge 2023.

Skills

Languages: TypeScript, JavaScript, Python, Java, C#, C++

Frontend: React 18/19, Next.js, Angular, Vite, Tailwind CSS, shadcn/ui, Radix UI, Chakra UI, Framer Motion, Storybook, Module Federation, Rspack

Backend: Node.js, Hono, Express, FastAPI, Spring Boot 3, ASP.NET Core, Entity Framework Core

Testing & Automation: Playwright, Vitest, Karma, Mocha, Chai, Sinon, axe-core, Selenium

Databases: MongoDB, PostgreSQL (Supabase), SQL Server, Firebase, Redis

DevOps / Cloud: AWS CodeArtifact, AWS S3, Azure, Cloudflare Workers, Docker, NGINX, GitHub Actions, Netlify

ML / NLP: TensorFlow, HuggingFace Transformers, BPE Tokenization, TextRank, TF-IDF, scikit-learn

Concepts: Design Systems, Micro-frontends, Dynamic Program Analysis, REST APIs, Microservices, OOP, Design Patterns, Agile/Scrum

Awards & Achievements

ICT Division Fellowship, Government of Bangladesh: research funding for NavA11y / M.Sc. thesis (Apr 2026)

Top 6, ICSE SCORE 2023 (international undergraduate SE competition) · **Champion**, Refactoring Competition (2023)

Best UI/UX, BUET Hackathon (2023) · **Champion**, Project Showcasing (2022) · **Runners-up**, Project Showcasing (2021)

600+ problems solved on Codeforces & LeetCode